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Modern Voucher Logistics

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Executive summary

We present a structured exploration of voucher-based economies with a focus on replicating vouchers, lineage formation, and trust networks. We formalize the underlying objects and relations, analyze their economic implications under zero-trust assumptions, and demonstrate why such systems resist exponential growth. We conclude with design directions for future voucher types.

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1. Voucher Economics

1.1. Voucher Basics

Definition 1.1 (Voucher). A **voucher** is an abstract object that can spawn at most one associated **token** upon invocation.

Definition 1.2 (Token). A **token** is a market good spawned by invoking a voucher, requiring payment of a fixed **token fee**. Ownership of the token is declared by the invocator.

Definition 1.3 (Cutoff). The **cutoff** is a global point in time after which ownership of multiple tokens is equivalent in value to ownership of a single token.

proposition 1.4 (Zero-Trust Ownership). *Authenticity of ownership transfers of vouchers and tokens cannot be guaranteed.*

As a consequence of ??, all economic reasoning in voucher systems must assume zero trust.

1.2. Birth of a Voucher Market

Definition 1.5 (Genesis Voucher). A **genesis voucher** is a voucher with no predecessor.

Definition 1.6 (Initial Pool). The **initial pool of vouchers** is the set of all genesis vouchers.

Genesis vouchers are initially controlled by a divine entity, which covertly delegates a subset of them to distributors.

1.3. Voucher Types

Definition 1.7 (End-Voucher). An **end-voucher** is a voucher with no replication mechanism.

Definition 1.8 (Replicating Voucher). A **replicating voucher** is a voucher capable of creating a successor voucher.

Definition 1.9 (Replication Delay). The **replication delay** is the (known or unknown) time between invocation of a voucher and delivery of its successor.

Definition 1.10 (Standard Voucher). A **standard voucher** is a replicating voucher that can create exactly one successor standard voucher.

Definition 1.11 (Angel Voucher). An **angel voucher** is a replicating voucher that can create exactly one end-voucher.

1.4. Voucher Lineage

Definition 1.12 (Predecessor and Successor). Let A and B be vouchers. If A created B , then A is the **predecessor** of B , written $\text{pred}(B) = A$, and B is a **successor** of A . If A is a standard voucher, its successor is unique and denoted $\text{succ}(A)$.

Definition 1.13 (Voucher Graph). Let V be the set of all vouchers. The **voucher graph** is the directed graph on V with edges $A \rightarrow B$ if $\text{pred}(B) = A$.

Lemma 1.14 (Acyclicity). *The voucher graph is a directed acyclic graph.*

Definition 1.15 (Lineage). The **lineage** of a voucher A is the set of all tokens associated with A and with any voucher B such that $B \prec A$, where $\prec$ is the transitive closure of the voucher graph.

1.5. Voucher Distribution

Definition 1.16 (Distributor). A **distributor** is an agent selected to receive vouchers from the initial pool.

Creation of a successor voucher is triggered by spawning the associated token, with both ownerships declared by the invocator.

2. The Voucher Economy

2.1. Voucher Utility

proposition 2.1 (Non-Accumulation of Financial Value). *Under zero-trust assumptions and after the cutoff, ownership of multiple tokens provides no additional financial value over a single token.*

Definition 2.2 (Social Value). **Social value** is a non-monetary utility derived from influence, prestige, or delegation power obtained through token and voucher distribution.

Lemma 2.3 (Dominance of Social Value). *In typical voucher economies, social value dominates direct financial value.*

2.2. Bottlenecks

Definition 2.4 (Human Delay). **Human delay** is additional latency in token spawning caused by capital constraints, coordination costs, or human error.

proposition 2.5 (Lineage Sensitivity). *Increased human delay reduces the maximal achievable lineage length before the cutoff.*

3. Trust Networks

Definition 3.1 (Trust Network). A **trust network** is a recursive distribution structure in which vouchers or tokens are transferred along social trust relationships.

proposition 3.2 (Lineage Maximization Strategy). *A voucher owner maximizes lineage by minimizing human delay and controlling successive ownership declarations.*

3.1. Pyramid Schemes

Definition 3.3 (Pyramid Scheme). A **pyramid scheme** is a distribution system with exponential growth in participant obligations or rewards.

proposition 3.4 (Trust Networks Are Not Pyramid Schemes). *Trust networks induced by standard vouchers are not pyramid schemes.*

The formal proof is deferred to section A.

4. Problems

4.1. Exploits

Definition 4.1 (Double-Sale Exploit). A **double-sale exploit** occurs when a malicious actor sells the same token ownership declaration to multiple parties.

proposition 4.2 (Exploit-Induced Pyramid Risk). *Double-sale exploits may locally induce pyramid-scheme-like behavior.*

5. Future Work

Definition 5.1 (Infinity Voucher). An **infinity voucher** is a proposed replicating voucher capable of spawning arbitrarily many non-replicating end-vouchers, subject to per-timeframe limits.

6. Conclusion

Voucher economies are governed less by monetary incentives than by formal constraints on replication, time, and trust. By modeling vouchers as a DAG with bounded growth and by identifying lineage as the core driver of social value, we explain both the resilience and the fragility of such systems. Future voucher types that remove trust assumptions while preserving bounded growth remain a promising direction.

A. Formal Properties of Trust Networks

Lemma A.1 (Tree Structure). *The subgraph of the voucher graph induced by standard vouchers is a forest of rooted trees.*

Lemma A.2 (Linear Growth). *The total number of vouchers and tokens created before the cutoff grows linearly in time.*

proposition A.3 (Anti-Exponential Propagation). *Trust networks formed from standard vouchers exhibit anti-exponential layer growth.*

proposition A.4 (Non-Pyramid Property). *Trust networks do not satisfy the defining properties of pyramid schemes.*

B. Simulation Model

Definition B.1 (Simulation Assumptions). *Simulations assume fixed token fees, bounded replication delays, rational agents, and social-value-maximizing behavior.*

proposition B.2 (Empirical Validation). *Under the simulation assumptions, observed voucher graphs are shallow, highly clustered, and consistent with the analytical results of the main text.*